Product Requirement Document (PRD)

(CSIT321 - Final Year Project)

CSIT-26-S2-05: A Mobile Application for Wise Workout

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1. Introduction

1.1 Overview

Wise Workout is an AI-enhanced fitness tracking and expert service platform designed to help users record workout activities, understand their progress, and connect with verified experts across different fitness and wellness categories. The platform combines workout data recording, progress analytics, AI-powered guidance, and expert services within a single mobile application.

The proposed product is intended for users who want to improve their fitness through self-guided tracking as well as users who prefer professional support from experts such as running coaches, cycling coaches, football coaches, strength trainers, nutritionists, and recovery specialists. By combining data-driven insights with expert-led services, Wise Workout aims to provide a more complete fitness support platform than a standard workout tracker.

1.2 Problem Statement

Many fitness users can record workout activities using existing applications, but they may still struggle to interpret what their data means and decide what actions to take next. Basic statistics such as calories burned, workout duration, and distance are useful, but they do not always provide clear guidance on whether the user is improving, undertraining, overtraining, or following a suitable plan.

In addition, users who require more specific support may need to search outside the application for professional advice. This creates a fragmented experience where workout tracking, progress interpretation, and expert guidance are separated across different platforms. Wise Workout addresses this problem by combining workout tracking, AI-powered progress interpretation, personalised recommendations, and access to verified experts in one platform.

1.3 Project Scope

The project scope focuses on developing a mobile application and supporting platform features for fitness tracking, AI-assisted guidance, expert service discovery, and system administration.

In scope:

- User registration, login, and role-based access control.
- Fitness profile and goal management for registered users.
- Workout recording for activities such as running, cycling, football, gym training, walking, and general exercise.
- Progress analytics, dashboards, and report summaries based on recorded workout data.
- AI-assisted progress summaries, workout recommendations, rest suggestions, and expert matching.
- Expert profiles, expert categories, service listings, and expert-created content.
- User discovery of experts and simple expert service request workflow.
- Free and premium feature access control.
- System admin features for managing users, expert applications, categories, listings, and platform content.
- Marketing website to explain the product and encourage registration.

Out of scope for the initial implementation:

- Real payment gateway integration.
- Full wearable device synchronisation.
- Video consultation and real-time expert chat.

- Medical diagnosis or clinical health advice.
- Advanced nutrition diagnosis or disease-specific diet planning.
- Large-scale social networking features.

1.4 Project Objectives

1. To design and develop a mobile platform that allows users to record and manage workout activities.
2. To provide users with progress analytics that help them understand workout consistency, calories burned, activity trends, and goal progress.
3. To implement AI-assisted features that summarise progress and recommend suitable workouts, rest actions, or expert categories.
4. To create an expert service platform where verified experts can list services, publish content, and support users across multiple fitness and wellness categories.
5. To support free and premium user tiers through role-based access and feature restrictions.
6. To provide administrative tools for managing users, expert applications, categories, service listings, and platform quality.
7. To produce a scalable project structure that can support future enhancements such as payments, wearable integrations, and live consultations.

2. Research Summary

2.1 Overview of Fitness App Market

The fitness app market includes applications that support activity tracking, workout planning, progress monitoring, nutrition tracking, social fitness, and digital coaching. Users increasingly expect fitness applications to provide not only raw activity data, but also personalised insights and convenient access to guidance.

For Wise Workout, the market research should focus on identifying how existing applications support workout tracking, analytics, AI or adaptive coaching, and access to professional fitness guidance. This section should be expanded later with verified market statistics and cited sources.

2.2 Fitness Applications in the Market

Application	Main Focus	Relevance to Wise Workout
Strava	Activity tracking, social fitness, running and cycling community features.	Strong community and performance tracking, but expert service access is not the central focus.
MyFitnessPal	Nutrition tracking, calorie logging, and health-related goal management.	Strong nutrition and diet logging, but not primarily an expert services marketplace.
adidas Running	Running and workout tracking with plans, challenges, and motivation features.	Good tracking and motivation, but expert marketplace features are limited.
Google Fit	General health and activity tracking through phone and wearable data.	Useful as a health data hub, but limited coaching and expert service functionality.
Freeletics	AI-supported fitness coaching and workout plans.	Strong personalised training focus, but less emphasis on multi-category expert services.
MapMyFitness	Workout tracking, route tracking, and activity history.	Good tracking coverage, but expert-led services are not the main platform model.
Lyfta	Workout logging, gym tracking, and fitness progression tools.	Strong workout recording focus, but limited expert service marketplace features.

2.3 Product Comparison Table

Feature / Function	Existing Apps	Wise Workout Target	MVP

Activity tracking	Yes	Yes	Yes
Workout progress dashboard	Yes	Yes	Yes
AI progress summary	Limited/varies	Yes	Yes
AI workout recommendation	Limited/varies	Yes	Yes
Expert profile discovery	Limited	Yes	Yes
Expert service request	Limited	Yes	Yes
Expert-created content	Limited	Yes	Yes
Free and premium access control	Yes	Yes	Yes
Admin expert approval	Limited	Yes	Yes

2.4 Key Findings from Existing Apps

- Most fitness applications support some form of workout or activity tracking.
- Many applications provide dashboards and charts, but users may still need help interpreting progress.
- AI or adaptive coaching is increasingly used to personalise fitness guidance.
- Expert-led services are often separate from workout tracking platforms.
- There is an opportunity to combine tracking, AI insights, and expert discovery into a single user experience.

2.5 Research Gap

The research gap identified for Wise Workout is the lack of a combined platform that allows users to track workouts, receive AI-assisted progress interpretation, and access verified experts from multiple fitness and wellness categories. Existing products may offer parts of this experience, but Wise Workout aims to combine these functions into one integrated platform.

2.6 Case Studies

Case studies can be developed later around existing fitness applications such as Strava, MyFitnessPal, Freeletics, and Google Fit. These case studies should compare how each product handles workout tracking, user engagement, personalisation, premium features, and expert guidance. The findings can then be used to justify Wise Workout's product direction.

2.7 Data Acquisition and Connectivity Protocols

For the MVP, workout data will primarily be acquired through manual user input within the mobile application. Users may record details such as workout type, date, duration, distance, calories burned, intensity, and notes. Optional device or wearable integration can be treated as a future enhancement.

Potential future data sources include smartphone sensors, wearable devices, and third-party health platforms. These integrations should be considered only after the core workout tracking and analytics features are stable.

2.8 Algorithms / Techniques Considered

The system may use a combination of rule-based logic and AI-assisted text generation. Rule-based logic can be used to evaluate workout frequency, goal progress, intensity balance, and recommendation triggers. AI can then convert these structured results into user-friendly explanations and suggestions.

Technique	Description	Purpose
Rule-based recommendation	Uses predefined conditions to recommend workouts, rest days, or expert categories.	Reliable, explainable, and suitable for MVP.
AI progress summarisation	Generates natural language summaries based on workout history and analytics.	Improves user understanding of fitness data.
AI expert matching	Suggests suitable expert categories or profiles based on user goals and activity patterns.	Connects the AI feature directly to the expert platform.

Analytics dashboard	Uses charts and summary statistics to display progress.	Provides visual evidence of improvement over time.

3. Target Users and User Categories

Wise Workout supports five main user categories: unregistered user, registered free user, registered premium tier, expert, and system admin. The expert role is not limited to one fitness category and may include specialists in running, cycling, football, strength training, nutrition, recovery, and other fitness-related areas.

User Category	Description	Main Responsibilities / Access
Unregistered User	A visitor who has not created an account.	View marketing website, read basic product information, browse limited public content, register or log in.
Registered Free User	A user with a free account.	Create fitness profile, log workouts, view basic progress dashboard, receive limited AI summaries, browse experts.
Registered Premium Tier	A paying or upgraded user.	Access advanced analytics, full AI recommendations, premium expert content, and service request features.
Expert	A verified service provider in a fitness or wellness category.	Create expert profile, list services, publish content, receive and manage service requests.
System Admin	Internal platform administrator.	Approve experts, manage users, manage categories, monitor content and listings, maintain platform quality.

3.1 Unregistered User

An unregistered user can access the marketing website, understand the purpose of Wise Workout, view public information about app features and expert services, and choose to register or log in. This user category supports user acquisition and first-time product discovery.

3.2 Registered Free User

A registered free user can create a basic fitness profile, record workout activities, view basic progress information, and browse available experts. Free users may receive limited AI-generated summaries or basic recommendations, but advanced analytics and premium expert features are restricted.

3.3 Registered Premium Tier

A registered premium user has access to enhanced features such as advanced progress analytics, full AI recommendations, AI expert matching, premium expert content, and expert service request functions. This tier supports the subscription-based business model.

3.4 Expert User

An expert user is a verified service provider who offers fitness or wellness guidance through the platform. Experts may belong to different categories such as running, cycling, football, strength training, nutrition, recovery, or general fitness. Experts can manage their profiles, publish content, and list services that users can discover or request.

3.5 System Admin

A system admin manages the operational side of the platform. Admins review expert applications, approve or reject expert profiles, manage expert categories, monitor service listings and content, manage users, and help maintain platform quality and trust.

4. Proposed Product

4.1 Product Motivation

The motivation behind Wise Workout is to provide a more complete digital fitness experience. Many users track their exercise data but do not always know how to interpret it or decide what to do next. At the same time, users who need professional support may have to rely on separate platforms to find coaches, trainers, or nutrition experts.

Wise Workout aims to reduce this gap by combining workout tracking, progress analytics, AI-powered guidance, and expert services in one mobile platform.

4.2 Proposed Product Overview

Wise Workout is an AI-enhanced fitness tracking and expert service platform. It allows users to record workout activities, monitor progress, receive AI-generated summaries and recommendations, and connect with verified experts across multiple fitness and wellness categories.

The platform supports both self-guided users and users who prefer professional guidance. Users can track workouts and receive AI-assisted insights, while experts can offer services, publish content, and provide specialised support based on their category of expertise.

4.3 Value Proposition

Stakeholder	Value Provided
For free users	A simple way to record workouts, track basic progress, and discover experts.
For premium users	Advanced analytics, AI-powered guidance, expert matching, and access to premium expert content or services.
For experts	A platform to promote fitness or wellness services, publish content, and reach users seeking specialised guidance.
For admins	Tools to manage platform quality, approve experts, and maintain a trusted service environment.

4.4 Main Features

Feature	Description
Authentication and Role Management	Registration, login, and feature access based on user role.
Fitness Profile Management	User goals, activity level, preferred workout categories, and basic physical information.
Workout Recording	Workout logs with type, duration, distance, calories, intensity, date, and notes.
Progress Analytics	Dashboard summaries, workout consistency, trends, and goal progress.
AI Fitness Assistant	AI-supported explanations and answers based on user fitness data.
AI Recommendation Engine	Workout, rest, and expert category suggestions based on user profile and activity history.
Expert Profiles	Expert biography, category, qualifications, services, and availability.
Expert Services	Service listings and simple user request workflow.
Expert Content	Training tips, workout guides, nutrition advice, and recovery content.
Subscription Access Control	Free and premium feature restrictions.
Admin Management	User, expert, category, service listing, and content management.
Marketing Website	Public website explaining features, value proposition, and expert services.

FR1: User Account and Role Management

The system shall allow users to register, log in, and access features based on their assigned role, including unregistered user, registered free user, registered premium user, expert, and system admin.

FR2: Fitness Profile Management

The system shall allow registered users to create and update their fitness profile, including fitness goals, activity level, preferred workout categories, and basic physical information.

FR3: Workout Recording and Activity Tracking

The system shall allow users to record workout activities, including workout category, duration, distance, calories burned, intensity level, date, and optional notes.

FR4: Progress Analytics and Reports

The system shall analyse recorded workout data and display progress summaries, trend charts, workout consistency, calories burned, and goal progress.

FR5: AI Progress Summary

The system shall generate AI-assisted natural language summaries that explain the user's workout progress, consistency, and recent performance trends.

FR6: AI Workout Recommendation

The system shall provide AI-assisted workout or rest recommendations based on the user's profile, goals, workout history, and recent activity level.

FR7: AI Expert Matching

The system shall recommend suitable expert categories or expert profiles based on the user's goals, preferred activities, and fitness support needs.

FR8: Expert Profile Management

The system shall allow verified experts to create and manage expert profiles, including expertise category, qualifications, biography, service description, and availability information.

FR9: Expert Service Listing Management

The system shall allow experts to create, update, and remove service listings such as training plans, coaching consultations, nutrition advice, and sport-specific guidance.

FR10: Expert Discovery and Service Request

The system shall allow users to search, filter, and view expert profiles or service listings, and submit a service request to an expert.

FR11: Expert Content Management

The system shall allow experts to publish fitness or wellness-related content, such as workout guides, training tips, nutrition advice, and recovery recommendations.

FR12: Subscription and Feature Access Control

The system shall control access to free and premium features, including advanced analytics, AI recommendations, premium content, and selected expert services.

FR13: System Administration

The system shall allow administrators to manage users, approve or reject expert applications, manage expert categories, monitor service listings, and maintain platform content.

FR14: Marketing Website Management

The system shall provide a public marketing website that introduces Wise Workout, explains its features, highlights expert services, and encourages users to register.

4.5 Marketing Website

The marketing website will act as the public entry point for Wise Workout. It should explain the platform's purpose, highlight key features, introduce expert services, compare free and premium access, and encourage users to register. It may include sections such as hero banner, feature overview, expert categories, pricing overview, frequently asked questions, and call-to-action buttons.

5. Business Model

5.1 Proposed Business Model

Wise Workout will use a freemium platform business model. Basic tracking features will be available to free users, while premium users will gain access to advanced analytics, expanded AI recommendations, premium expert content, and expert service request features. The expert platform can also support future revenue through expert listing fees, commission from service bookings, and featured expert promotions.

5.2 User Tiers

Tier / Role	Access
Free User	Basic fitness tracking, basic dashboard, limited AI summary, expert browsing.
Premium User	Advanced analytics, full AI recommendations, AI expert matching, premium expert content, and service requests.
Expert	Expert profile, service listings, content publishing, and user request management.
Admin	Platform management, expert approval, category management, and monitoring.

5.3 Free vs Premium Feature Access

Feature	Free User	Premium User
Workout logging	Yes	Yes
Basic fitness profile	Yes	Yes
Basic dashboard	Yes	Yes
Advanced analytics	No	Yes
AI progress summary	Limited	Full access
AI workout recommendation	Limited	Full access
AI expert matching	Basic	Advanced
Browse expert profiles	Yes	Yes
Submit expert service request	Limited / No	Yes
Premium expert content	No	Yes

5.4 Revenue Model

Revenue Source	Description
Premium subscription	Users pay for advanced analytics, full AI recommendations, and premium expert content.
Expert service commission	The platform may take a commission from completed expert service requests in future versions.
Expert listing fee	Experts may pay to list services or unlock enhanced visibility.
Featured expert promotion	Experts may pay for promoted placement in search or category pages.
Premium content access	Users may pay to access specialised expert-created guides or plans.

5.5 Justification of Business Model

The freemium model is suitable because it allows users to try the core workout tracking features before upgrading for deeper value. Premium access is justified through advanced analytics, AI-powered guidance,

expert matching, and expert content. The expert service model also creates a platform ecosystem where experts gain visibility and users gain access to specialised support.

6. Functional Hierarchy Diagram

The functional hierarchy below shows the main platform modules and their related functions. This can later be converted into a formal diagram.

Wise Workout Platform
- Authentication and Role Management
  - Register account
  - Log in / log out
  - Manage role-based access
- User Fitness Profile Management
  - Create profile
  - Update goals
  - Manage preferences
- Workout Recording and Activity Tracking
  - Add workout
  - View workout history
  - Edit or delete workout
- Progress Analytics and Reports
  - View dashboard
  - View weekly summary
  - View trend reports
- AI Features
  - Generate progress summary
  - Recommend workout or rest
  - Match user to expert category
- Expert Platform
  - Create expert profile
  - Manage service listings
  - Publish content
  - Manage user requests
- Subscription Access Control
  - Manage free access
  - Manage premium access
- Admin Management
  - Manage users
  - Approve experts
  - Manage categories
  - Monitor content and listings
- Marketing Website
  - View product information
  - View feature highlights
  - Register or log in

7. Use Case Diagram and Description

7.1 User Stories

Actor	User Story
Unregistered User	As an unregistered user, I want to view the marketing website so that I can understand the app before registering.
Registered Free User	As a free user, I want to log my workouts so that I can track my fitness activity.
Registered Free User	As a free user, I want to view a basic dashboard so that I can understand my recent progress.
Registered Premium User	As a premium user, I want to receive AI workout recommendations so that I can decide what to do next.
Registered Premium User	As a premium user, I want to be matched with suitable experts so that I can get relevant professional support.
Expert	As an expert, I want to create a profile so that users can discover my services.
Expert	As an expert, I want to publish fitness content so that I can share guidance with users.
Expert	As an expert, I want to manage service requests so that I can respond to interested users.
System Admin	As an admin, I want to approve expert applications so that only verified experts are listed on the platform.
System Admin	As an admin, I want to manage categories so that expert services are organised clearly.

7.2 Preliminary Use Case Diagram

The preliminary use case diagram should include five actors: Unregistered User, Registered Free User, Registered Premium User, Expert, and System Admin.

Actor	Use Cases
Unregistered User	View marketing website, Register account, Log in
Registered Free User	Manage fitness profile, Log workout, View basic dashboard, Browse experts
Registered Premium User	View advanced analytics, Receive AI recommendation, Receive AI expert matching, Submit expert service request, Access premium content
Expert	Apply as expert, Manage expert profile, Manage service listing, Publish content, Respond to service request
System Admin	Manage users, Approve experts, Manage expert categories, Monitor content, Manage platform settings

7.3 Use Case Descriptions

ID	Use Case	Primary Actor	Description	Outcome
UC01	Register Account	Unregistered User	User creates an account to access platform features.	User account is created.
UC02	Log Workout	Registered User	User records workout details such as type, duration, distance, intensity, and notes.	Workout record is saved.
UC03	View Progress Dashboard	Registered User	User views workout summaries, trends, and goal progress.	Dashboard is displayed.
UC04	Generate AI Recommendation	Premium User	System analyses profile and workout history to suggest workout, rest, or expert category.	Recommendation is displayed.
UC05	Manage Expert Profile	Expert	Expert creates or updates profile and expertise information.	Expert profile is saved.
UC06	Submit Service Request	Premium User	User submits a request to an expert service listing.	Request is sent to expert.
UC07	Approve Expert Application	System Admin	Admin reviews and approves or rejects expert applications.	Expert status is updated.

8. Project Schedule

8.1 Project Timeline

Phase	Name	Main Activities
Phase 1	Planning and Requirements	Finalise PRD, define user roles, confirm features, produce diagrams.
Phase 2	UI/UX and System Design	Create wireframes, database design, architecture, and use case documentation.
Phase 3	Core Development	Implement authentication, profiles, workout logging, dashboard, and admin base.
Phase 4	AI and Expert Platform	Implement AI summaries, recommendations, expert profiles, service listings, and request workflow.
Phase 5	Testing and Refinement	Conduct functional testing, usability testing, bug fixing, and documentation updates.
Phase 6	Final Delivery	Prepare final report, presentation, demonstration, and project handover.

8.2 Gantt Chart

The Gantt chart should be produced after the final feature list is confirmed. It should map each phase to project weeks and assign tasks to team members. The chart should include milestones such as PRD completion, UI prototype completion, core feature completion, AI feature completion, expert platform completion, testing completion, and final submission.

9. Development Tools

9.1 Summary of Development Tools

Area	Tool / Technology	Purpose
Frontend Mobile App	Flutter / React Native	Cross-platform mobile development for iOS and Android.
Backend API	Node.js / Express or similar	Provides APIs for users, workouts, experts, AI requests, and admin features.
Database	PostgreSQL / MySQL / Firebase	Stores users, workout logs, expert profiles, content, requests, and subscriptions.
AI Integration	LLM API or AI service	Generates progress summaries, recommendations, and expert matching explanations.
Design Tool	Figma	Creates wireframes and UI prototypes.
Version Control	GitHub	Tracks source code changes and supports team collaboration.
Testing	Manual testing and API testing tools	Validates app functions, API responses, and user workflows.

9.2 Justification of Development Tools

The selected tools should support fast development, maintainability, and cross-platform delivery. A cross-platform mobile framework is suitable because the application is intended for mobile users. A backend API separates business logic from the mobile interface and supports future scaling. A structured database is required to manage user accounts, workout records, expert profiles, service listings, and AI-related outputs.

10. Development Methodologies

10.1 Agile Iterative Development Methodology

The project will follow an Agile iterative development methodology. This approach is suitable because the product includes multiple connected modules, such as workout tracking, AI recommendations, expert services, and admin management. The team can build and test the system in increments, gather feedback, and refine features progressively.

Each iteration may focus on a specific module or group of related features. For example, early iterations can focus on authentication and workout logging, while later iterations can focus on AI recommendations, expert services, and admin controls.

The Agile approach supports continuous improvement and allows the team to adjust the project scope if implementation complexity increases.

11. References

References will be added after the research summary and market analysis are finalised with verified sources. Suggested reference areas include fitness app market reports, official product websites of comparison apps, AI in fitness applications, and mobile health platform design.